Autonomous Career Advisor

Multi-Agent System for Skill Gap Analysis and Learning Path Generation

Project Review – 1 : Reference Document

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1. Project Goal & Motivation

Students today pick courses and learning paths blindly, with no visibility into what skills the job market actually demands. This project builds an AI-powered career guidance system that connects a student's real skill profile to real-time industry demand, and generates a personalized, adaptive learning path to close that gap.

Goal

- Develop an AI-powered career guidance system using multiple intelligent agents.
- Analyse a user's current skills and identify missing competencies.
- Generate personalized learning paths based on individual career goals.
- Recommend industry-relevant skills using current job market demand.
- Help students choose the right career and learning path.

Alignment — SDG 4: Quality Education

- Personalized learning experience.
- AI-powered career guidance.
- Industry-relevant skill development.
- Better education-to-employment pathway.

2. Existing System vs. Proposed Idea

Existing System	Proposed System
Generic course recommendations, content-delivery focused.	Multi-agent AI framework connecting education with real industry requirements.
No personalized skill-gap analysis.	Dedicated Skill Gap Analyzer comparing student vs. market skills.
Ignores current industry demand.	Live Market Demand Agent pulling real-time job market data.
Static learning paths, minimal personalization.	Adaptive learning paths that re-plan as the student progresses.

Differentiation from Base / Related Papers

Base paper — EduPlanner (Zhang et al., IEEE TLT 2025) proposes an LLM-based multi-agent system for instructional design, but has zero career or job-market coverage. This project extends the multi-agent LLM planning idea from the classroom into career guidance: a live market-demand agent (RAG over job APIs), an ESCO/O*NET-based skill taxonomy for the gap engine, a verifier/critic agent, and an adaptive re-planning loop.

Closest related work — GenMentor (WWW'25 companion) is the nearest prior art in agentic career mentoring; it is cited and differentiated, not used as the base paper. This project's differentiators over GenMentor are the live market-data RAG pipeline and the explicit skill-gap + verifier loop.

3. Technology Stack

Layer	Technology	Purpose
Agent orchestration	LangGraph (LangChain)	Coordinates the specialized agents and manages state/handoffs between them
Language model	GPT-4 class LLM	Powers reasoning inside each agent — skill matching, recommendation generation
Retrieval (RAG)	Vector DB — Chroma / FAISS	Stores embeddings of skills, courses and job postings for similarity search
Market data source	Job APIs — Adzuna / JSearch	Live job postings feed the Market Demand Agent instead of static data
Skill taxonomy	ESCO / O*NET	Standardized skill/occupation vocabulary so synonymous skills map to one node
Skill-gap matching	Embedding similarity + rule-based comparison	Compares student skill vectors against required skill vectors
Frontend / dashboard	Streamlit	Interactive UI for profile input and roadmap display
Evaluation	LLM-as-judge + verifier/critic agent	Sanity-checks recommendations before they reach the student

Note: this stack reflects the base-paper analysis and is pending final mentor confirmation before implementation (Weeks 7–10 of the project timeline).

4. Module Breakdown

Pipeline: profile in → Student Agent normalizes it → Market Agent fetches live demand → Gap Analyzer diffs the two → Path Generator plans the roadmap → Dashboard shows it.

Module 1 — User Profile Management

Front door of the system. Collects student input (education, current skills, career interest) and stores it as a structured profile.

Tech: Streamlit form → profile stored in DB / session state.

Module 2 — Student Profile Agent

Reads the raw profile and normalizes skill names against the ESCO/O*NET taxonomy, then estimates a proficiency level per skill (beginner / intermediate / advanced) rather than a simple yes/no list.

Tech: LLM reasoning + taxonomy lookup.

Module 3 — Market Demand Agent

Pulls what the job market currently wants — in-demand skills, trending roles, salary bands — via live job-posting APIs instead of a static hardcoded list. This is the project's key novelty over the base paper.

Tech: RAG pipeline: API fetch → embed → store in vector DB.

Module 4 — Skill Gap Analyzer

Takes Module 2's output (what the student has) and Module 3's output (what the market wants) and computes the delta — which skills are missing, which are weak.

Tech: Embedding similarity search over the vector DB + comparison logic.

Module 5 — Learning Path Generator

Takes the gap list and builds an ordered roadmap — which skill to learn first, what course/project covers it. The path is adaptive: as the student progresses, it can re-plan rather than staying static.

Tech: LLM generates the plan; a verifier/critic agent checks it is coherent and achievable before it is shown.

Module 6 — Dashboard & Recommendation

Output layer — displays the roadmap, tracks learning progress, and surfaces career recommendations.

Tech: Streamlit dashboard reading from the profile + roadmap state.

5. Expected Results & Advantages

Expected Results

- Accurate identification of user skill gaps.
- Personalized learning roadmap for every student.
- Career recommendations aligned with industry requirements.
- Improved learning efficiency through structured guidance.

Advantages

- Saves time by focusing on essential skills.
- Enhances employability using market-driven recommendations.
- Supports continuous learning and career development.
- Improves decision-making through AI-powered insights.

6. Presentation Walkthrough (Talking Points)

Opening

State it plainly: an AI system that analyzes skill gaps against real job-market demand and generates a personalized learning path. One sentence, no fluff.

Goal & SDG

Frame as problem → solution: students pick courses blindly today; the goal is to connect current skills, market demand, and a generated path.

Existing System

Present as contrast, not a list — existing platforms give generic, static recommendations with no gap analysis or market awareness.

Proposed Idea

Emphasize what's new: agent specialization + live market data + adaptive (not static) paths.

Proposed System / Architecture diagrams

Walk the diagram left to right live — point at boxes, don't read text off the slide.

Algorithms

Group into two ideas: how agents talk to each other (LangGraph, LLM) vs. how recommendations are made (RAG, gap algorithm, recommendation algorithm).

Modules

Narrate as a pipeline story (profile → student agent → market agent → gap analyzer → path generator → dashboard), not a static list.

References

Be ready to explain the EduPlanner vs. GenMentor differentiation — see Section 2 above.

Delivery Tips

- Don't read slides verbatim — use them as prompts, explain in your own words.
- Since Review-1 is pre-implementation, spend more time justifying the problem (goal, existing-system gap) than on algorithms not yet built.

- Have one ready answer for "why multi-agent instead of a single LLM call": each agent has a distinct data source/responsibility (student data vs. live market data vs. planning logic) that a single monolithic prompt can't cleanly separate or update independently.

7. References

1. Zhang, X., Zhang, C., Sun, J., Xiao, J., Yang, Y., & Luo, Y., "EduPlanner: LLM-Based Multiagent Systems for Customized and Intelligent Instructional Design," IEEE Transactions on Learning Technologies, vol. 18, pp. 416–427, 2025. (Base paper)
2. Wang, X., et al., "A Second-Classroom Personalized Learning Path Recommendation System Based on Large Language Model Technology," Applied Sciences, vol. 15, no. 14, 2025. (Fallback base paper)
3. "Artificial Intelligence in Personalized Learning: A Global Systematic Review of Current Advancements," Computers & Education, Elsevier, 2025.
4. Bardach, L., et al., "Intelligent Tutoring Systems Need Teachers," Journal of Computer Assisted Learning, Wiley, 2025.
5. Ouyang, Z., "Self-Regulated Learning and Engagement as Serial Mediators Between AI-Driven Adaptive Learning Platform Characteristics and Educational Quality," Frontiers in Psychology, 2025.